Joint Distributions

Joint PMF (discrete):

$$p_{X,Y}(x,y) = P(X = x, Y = y)$$

$$\sum_x \sum_y p_{X,Y}(x,y) = 1$$

Joint PDF (continuous):

$$f_{X,Y}(x,y)$$

$$\int \int f_{X,Y}(x,y)\,dx\,dy = 1$$

Marginal Distributions
How to find marginals from joint:
Discrete:

$$p_X(x) = \sum_y p_{X,Y}(x,y)$$

$$p_Y(y) = \sum_x p_{X,Y}(x,y)$$

Continuous:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$$

—

Conditional Distributions

Definition (discrete):

$$p_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)}$$

Definition (continuous):

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

How to find conditional density step-by-step:
1. Start with joint density $f_{X,Y}(x,y)$ 2. Find marginal $f_X(x)$ by integrating out y 3. Divide:

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

Similarly,

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

—

Independence

X and Y are independent if:
Discrete:

$$p_{X,Y}(x,y) = p_X(x)p_Y(y)$$

Continuous:

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

If independent:

$$E[XY] = E[X]E[Y]$$

—

Expectation

Discrete:

$$E[X] = \sum_x x p_X(x)$$

Continuous:

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx$$

Function of RV:

$$E[g(X)] = \sum g(x)p_X(x)$$

$$E[g(X)] = \int g(x)f_X(x)\,dx$$

Linearity of Expectation

$$E[aX + bY + c] = aE[X] + bE[Y] + c$$

—

Variance

Definition:

$$\mathrm{Var}(X) = E[(X - \mu)^2]$$

Simplified:

$$\mathrm{Var}(X) = E[X^2] - \mu_x^2$$

Shortcut Formula:

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

$$\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$$

If independent:

$$\mathrm{Var}(aX+bY+c) = a^2\mathrm{Var}(X)+b^2\mathrm{Var}(Y)$$

—

Covariance

Definition:

$$\mathrm{Cov}(X,Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Computational Formula:

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

Simplified:

$$\mathrm{Cov}(X,Y) = E[XY] - \mu_x\mu_y$$

General form (not independent):

$$\mathrm{Var}(aX \pm bY) = a^2\mathrm{Var}(X)+b^2\mathrm{Var}(Y)\pm 2ab\,\mathrm{Cov}(X$$

If independent:

$$\mathrm{Cov}(X,Y) = 0$$

—

Correlation

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}$$

$$-1 \le \rho \le 1$$

—

Conditional Expectation

$$E[Y|X = x] = \sum_y y\,p_{Y|X}(y|x)$$

$$E[Y|X = x] = \int y\,f_{Y|X}(y|x)\,dy$$

Law of Total Expectation

$$E[Y] = E[E[Y|X]]$$

—

Law of Total Variance

$$\mathrm{Var}(Y) = E[\mathrm{Var}(Y|X)] + \mathrm{Var}(E[Y|X])$$

—

Transformations of Random Variables

If $Y = g(X)$ and g is monotone:

$$f_Y(y) = f_X(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right|$$

Expectation and Variance under transformation

$$E[aX + b] = aE[X] + b$$
$$\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$$

—

CDF and PDF Relationship

$$F_X(x) = P(X \le x)$$

If continuous:

$$f_X(x) = \frac{d}{dx} F_X(x)$$

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$$

—

Sampling Distributions

Let $X_1, \ldots, X_n$ i.i.d.
Sample Mean:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$E[\bar{X}] = \mu, \quad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad SD(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

If population normal:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Sample Proportion:

$$E[\hat{p}] = p, \quad SD(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

—

Variance of Diff. of Sample Means (Dependent)

Let $\bar{X}_1, \bar{X}_2$ be sample means from samples of size n_1, n_2.
General (dependent case):

$$\mathrm{Var}(\bar{X}_1 \pm \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \pm 2\,\mathrm{Cov}(\bar{X}_1, \bar{X}_2)$$

Covariance of sample means:

$$\mathrm{Cov}(\bar{X}_1, \bar{X}_2) = \frac{1}{n_1 n_2}\sum_{i=1}^{n_1}\sum_{j=1}^{n_2} \mathrm{Cov}(X_i, Y_j)$$

Special case (paired samples, $n_1 = n_2 = n$):

$$\mathrm{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n} - 2\frac{\mathrm{Cov}(X_1, X_2)}{n}$$

Sample SD of differences (s_D) shortcut:

$$s_D = \sqrt{s_X^2 + s_Y^2 - (2 \cdot r \cdot s_X \cdot s_Y)}$$

If independent:

$$\mathrm{Cov}(\bar{X}_1, \bar{X}_2) = 0 \implies \mathrm{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

—

Central Limit Theorem

For mean (large $n \ge 30$):

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0,1)$$

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

For proportion:

$$\hat{p} \approx N\left(p, \frac{p(1-p)}{n}\right)$$

(valid if $np \ge 15$, $n(1-p) \ge 15$)

—

Empirical Estimators

Sample mean:

$$\bar{x} = \frac{1}{n}\sum x_i$$

Sample variance:

$$s^2 = \frac{1}{n-1}\sum (x_i - \bar{x})^2$$

Sample covariance:

$$s_{XY} = \frac{1}{n-1}\sum (x_i - \bar{x})(y_i - \bar{y})$$

Sample correlation:

$$r = \frac{s_{XY}}{s_X s_Y}$$

—

Bayes' Theorem (General Form)

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Continuous:

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x) f_X(x)}{f_Y(y)}$$

—
—

Standard Error (SE)

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} \quad \text{or} \quad \frac{S}{\sqrt{n}}$$

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

—

Confidence Intervals (General)

$$CI = \text{Estimator} \pm (\text{Critical Value})(SE)$$

Margin of Error:

$$MoE = (\text{Critical Value})(SE)$$

A $100(1-\alpha)\%$ CI satisfies:

$$P(L(X_1, \ldots, X_n) \le \theta \le U(X_1, \ldots, X_n)) = 1 - \alpha$$

Interpretation: In repeated sampling, $(1-\alpha)$ proportion of intervals contain θ.

—

Specific CIs

CI for Mean (σ known):

$$\bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

CI for Mean (σ unknown):

$$\bar{X} \pm t_{\alpha/2, n-1}\frac{S}{\sqrt{n}}$$

CI for Proportion:

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Key Notes:

- CI length $= 2 \times MoE$
- Larger $n \Rightarrow$ smaller CI
- Higher confidence $\Rightarrow$ wider CI
- CLT requires large n or normal population

—

Sample Size for Estimating p

Solving for length L of $(1-\alpha)100\%$ CI:

$$n = \frac{4z_{\alpha/2}^2 \hat{p}(1-\hat{p})}{L^2}$$

(*Round up to next integer. $\hat{p}$ often unknown before sample.*)
If prior $p = p_0$ is known:

$$n = \frac{4z_{\alpha/2}^2 p_0(1-p_0)}{L^2}$$

If no prior info on p is available:

$$n = \frac{z_{\alpha/2}^2}{L^2}$$

Independence & Conditional Properties

Additional Independence Facts:
If X, Y are independent ($X \perp Y$):

- $f_{Y|X}(y|x) = f_Y(y)$ (conditional = marginal)
- $E[Y|X] = E[Y]$
- $Var(Y|X) = Var(Y)$

$\Rightarrow$ Conditional distribution does **NOT** depend on X.
Functions of independent RVs:
If $X \perp Y$, then $g(X) \perp h(Y)$.
Uncorrelated vs Independent:

- Independence $\Rightarrow Cov(X,Y) = 0$
- $Cov(X,Y) = 0$ does **NOT** $\Rightarrow$ independence
- *Counterexample:* $X \sim N(0,1), Y = X^2$

Equivalent Conditions for Independence:

- Joint PMF/PDF: $f_{X,Y}(x,y) = f_X(x)f_Y(y)$
- Joint CDF: $F_{X,Y}(x,y) = F_X(x)F_Y(y)$
- Probabilities: $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$

Conditional Expectation (General Function):

$$E[g(X)|Y=y] = \sum_x g(x)p_{X|Y}(x|y)$$

$$\text{or} \quad \int g(x)f_{X|Y}(x|y)\,dx$$

$E[g(X)|Y]$ is a function of $Y \Rightarrow$ random variable.

Covariance & Correlation (Advanced)

Covariance Properties:

- $Cov(X,c) = 0$
- $Cov(aX + b, cY + d) = ac\,Cov(X,Y)$

Correlation:

$$\rho = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

Properties:

- Unit-free
- $-1 \le \rho \le 1$

Effect of Scaling:
$Corr(aX, bY)$ is:

- Same sign as ρ if $ab > 0$
- Flips sign of ρ if $ab < 0$

Perfect Correlation:

- $\rho = 1 \iff Y = aX + b, a > 0$
- $\rho = -1 \iff Y = aX + b, a < 0$

Sample Correlation:

- Unit-free
- $-1 \le r \le 1$

Normal Distribution Moments

If $Z \sim N(0,1)$:
Odd moments: $E[Z^{2k+1}] = 0$
First four moments:

- $E[Z] = 0$
- $E[Z^2] = 1$
- $E[Z^3] = 0$
- $E[Z^4] = 3$

Conditional Variance

Definition:

$$Var(Y|X=x) = E[(Y - E[Y|X=x])^2|X=x]$$

Shortcut:

$$Var(Y|X) = E[Y^2|X] - (E[Y|X])^2$$

Law of Iterated Expectation (Product):

$$E[XY] = E[XE[Y|X]]$$

Sampling & Estimator Properties

Bias:

$$Bias(\hat{\theta}) = E[\hat{\theta}] - \theta$$

Unbiased:

- $E[\bar{X}] = \mu$
- $E[\hat{p}] = p$

Standard Error:

- $SE(\bar{X}) = \sigma/\sqrt{n}$ (σ known)
- $SE(\bar{X}) = s/\sqrt{n}$ (σ unknown)
- $SE(\hat{p}) = \sqrt{p(1-p)/n}$ (testing)
- $SE(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n}$ (CI)

CLT for proportions: $np \ge 15, n(1-p) \ge 15$

Confidence Intervals (Deeper)

Interpretation:

- **Before sampling:** $P(\theta \in CI) = 1 - \alpha$
- **After sampling:** θ is fixed $\Rightarrow$ probability is 0 or 1
- **Long-run:** $(1-\alpha)\%$ of intervals contain θ

Trade-off:

- Higher confidence $\Rightarrow$ wider interval
- Larger $n \Rightarrow$ narrower interval

Sample Size for Mean (σ known):

$$n = \left(\frac{z_{\alpha/2}\sigma}{MoE}\right)^2 \quad \text{(Round UP)}$$

Hypothesis Testing Framework

Hypotheses: H_0 (null), H_a (alternative)
Test Statistic:

$$\frac{\text{statistic} - \text{null value}}{SE}$$

p-value: $P(\text{result as extreme} \mid H_0 \text{ true})$
Decision Rule:

- $p \leq \alpha \Rightarrow$ Reject H_0
- $p > \alpha \Rightarrow$ Fail to reject H_0

Specific Tests

Z-test (mean, σ known): $Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$
T-test (σ unknown): $T = \frac{\bar{X}-\mu_0}{s/\sqrt{n}}$, $df = n-1$
Z-test (proportion): $Z = \frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}}$

Errors & Power

Type I Error: Reject H_0 when true.
Type II Error: Fail to reject H_0 when false.
Power: $Power = 1-\beta = P(\text{reject } H_0 \mid H_0 \text{ false})$

Paired Data

Let $D = X_1 - X_2$:

- $E[D] = \mu_D$
- $Var(D) = \sigma_1^2 + \sigma_2^2 - 2Cov(X_1, X_2)$
- $s_D^2 = \frac{\sum_{i=1}^n (D_i - \bar{D})^2}{n-1}$, where $D_i = X_{1,i} - X_{2,i}$

Paired t-test: $T = \frac{\bar{D}-\mu_{D,0}}{s_D/\sqrt{n}}$, $df = n-1$

Sample Size for Proportion

With estimate $\hat{p}$:

$$n = \frac{z_{\alpha/2}^2 \hat{p}(1-\hat{p})}{MoE^2}$$

Conservative (no info):

$$n = \frac{z_{\alpha/2}^2}{4MoE^2}$$

(Always **ROUND UP**)

Ultra-Condensed Logic & Edge Cases

Independence Checklist:

- If $f_{Y|X}(y|x)$ depends on $x \Rightarrow$ **NOT** independent
- If $E[Y|X]$ depends on $X \Rightarrow$ **NOT** independent
- If $Var(Y|X)$ depends on $X \Rightarrow$ **NOT** independent

Fast test: Check if $f_{X,Y}(x,y) = f_X(x)f_Y(y)$.

Covariance & Expectation Tricks:

- $E[aX + bY] = aE[X] + bE[Y]$
- $Cov(X,Y) = 0 \implies$ independence (unless jointly normal)
- If (X,Y) jointly normal: $Cov(X,Y) = 0 \Rightarrow$ independence

---

Conditional Expectation Tricks:
Tower Property (extended):

$$E[E[g(X,Y)|X]] = E[g(X,Y)]$$

Special case: $E[E[Y|X]] = E[Y]$

Variance Decomposition Insight:
Law of Total Variance:

$$Var(Y) = E[Var(Y|X)] + Var(E[Y|X])$$

Interpretation: Total variance = "within variability" + "between variability"

Hypothesis Testing Shortcuts

Test Statistic Pattern:

$$\text{Test Stat} = \frac{\text{Estimator} - \text{Null}}{SE} = \frac{\text{signal}}{\text{noise}}$$

Rejection Regions (Z/T):

- **Right-tailed:** reject if $stat > critical$
- **Left-tailed:** reject if $stat < -critical$
- **Two-sided:** reject if $|stat| > critical$

p-value shortcuts:

- **Right tail:** $P(Z > z)$
- **Left tail:** $P(Z < z)$
- **Two-sided:** $2P(Z > |z|)$

Confidence Interval Shortcuts

CI $\leftrightarrow$ Test Link: Reject $H_0 : \mu = \mu_0 \iff \mu_0 \notin CI$
Margin of Error (MoE):

$$MoE = (\text{critical value}) \times (SE)$$

$$\text{CI width} = 2 \times MoE$$

Quick intuition:

- Bigger $n \Rightarrow \downarrow SE \Rightarrow$ tighter CI
- Bigger $\sigma \Rightarrow \uparrow SE \Rightarrow$ wider CI

Proportion Nuances

Testing vs CI:

- **Test** uses p_0 in $SE = \sqrt{p_0(1-p_0)/n}$
- **CI** uses $\hat{p}$ in $SE = \sqrt{\hat{p}(1-\hat{p})/n}$

Reason: The test assumes H_0 is true.

Worst-case variance: $p(1-p)$ is maximized at $p = 0.5$. $\Rightarrow$ Conservative n uses 0.25 in the numerator.

Paired Data Insight

Variance of the Difference ($D = X_1 - X_2$):

$$Var(D) = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$$

- $\rho > 0 \Rightarrow$ reduces variance $\Rightarrow$ tighter CI
- $\rho < 0 \Rightarrow$ increases variance

Final Exam Strategy

Common Mistakes to Avoid:

- Using $\hat{p}$ instead of p_0 in hypothesis tests.
- Forgetting $df = n - 1$ in t-tests.
- Forgetting the factor of 2 in two-sided p-values.
- Forgetting to **round UP** for sample size n.

R Probability Functions: {prefix}{suffix}

- **Prefixes:**
  - `d`: Density (PDF/PMF height) [1, 2]
  - `p`: Cumulative Probability ($P(X \leq q)$) [1, 2]
  - `q`: Quantile/Inverse CDF (finds x given p) [1, 2]
  - `r`: Random Generation [3]
- **Suffixes (Distributions):**
  - `norm(q, mean, sd)`: Normal Distribution [2, 4]
  - `t(q, df)`: Student's t Distribution [5, 6]
  - `binom(x, size, prob)`: Binomial Distribution [7]
  - `unif(q, min, max)`: Uniform Distribution [2, 7]
  -
  - **Critical Value (Z):** `qnorm(1 - alpha/2)` for two-sided [6].
  - **Critical Value (T):** `qt(1 - alpha/2, df)` for two-sided [7].
- **Descriptive & Utility:**
  - `mean()`, `sd()`, `var()`, `IQR()`, `quantile(x, p)` [3, 5, 8]
  - `summary()`, `table()`, `prop.table()`, `round(val, n)` [1-3, 9]
  - `rowSums()`, `colSums()`, `sum()` [10, 11]